

FORENSIC EXAMINATION REPORT

Case 001-domexuser

Evidence: nps-2009-domexusers

Examination Platform: Autopsy

Report Type: Digital Forensic Examination

Examiner: Basirat A.Akintunde

Evidence Hash Verification

Source: FTK Imager/ Windows PowerShell

Evidence Type: Forensics Image

Verification result: Match

What it shows: The forensics image was successfully verified using FTK Imager. The completed MD5 and SHA-1 hashes matched the stored verification hashes, confirming the integrity of the acquired forensics image. No bad blocks were identified in the image.

```
PS C:\Users\basir\OneDrive\Desktop\forensics\Case-001-DomexUsers\Original-Evidence> Get-Childitem -File | Where-Object {
$.Name -like "nps-2009-domexusers (1).E01"} | Get-FileHash -Algorithm SHA256
```

Algorithm	Hash	Path
SHA256	5C52F16EDDD6D1AFE216D968B19E7267FBD5E3C8BB1626BFB2D8C4F36CFAA1C	C:\Users\basir\OneDrive\Deskt...

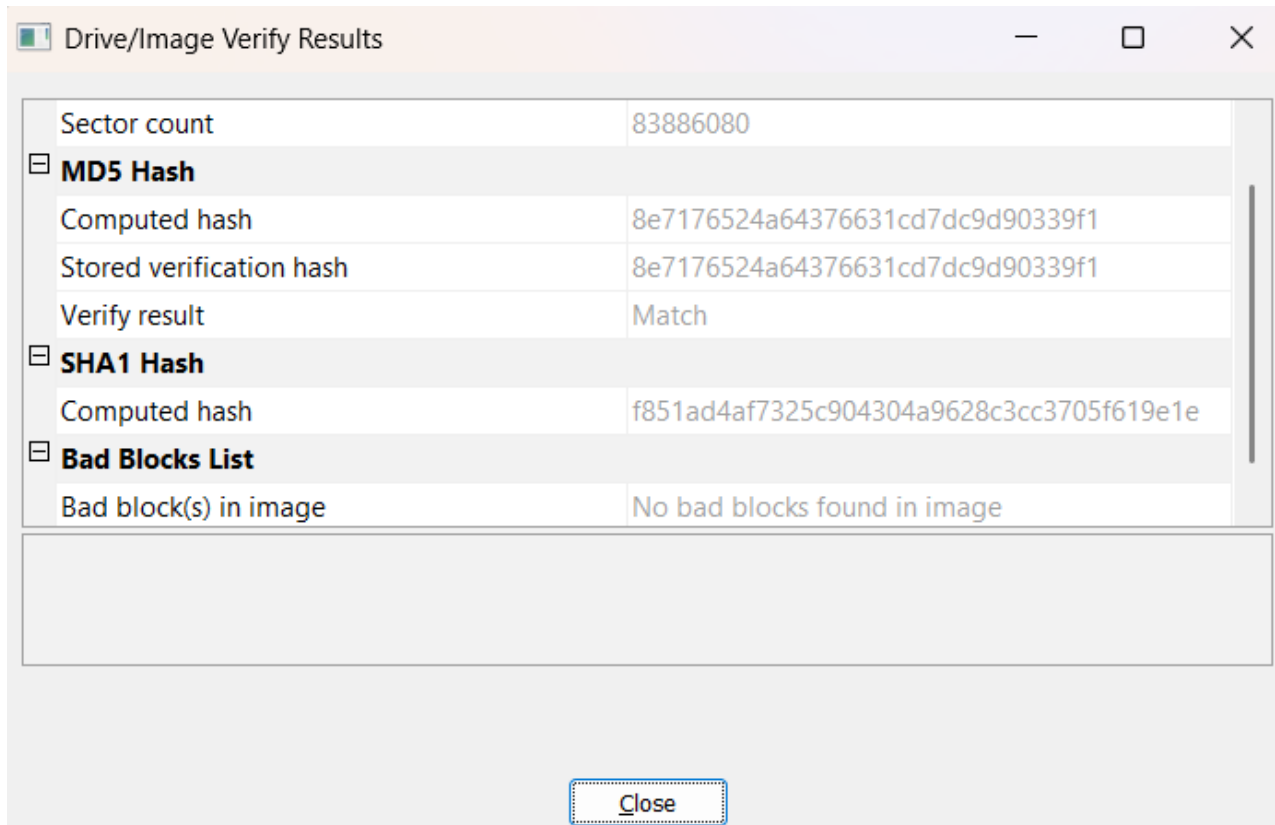


Figure 0.1–0.2. Forensic image hash verification results in FTK Imager showing matching MD5 and SHA-1 hashes and no bad blocks.

Artifact 1 – User Account

User: Administrator, domex1, domex2

Source: OS Accounts / Windows Registry

What it shows: The forensic image contains three local Windows user accounts: administrator, domex1, and domex2. These accounts are associated with the Windows system represented in the forensic image. The presence of an account does not by itself establish that the account was actively used during the relevant period.

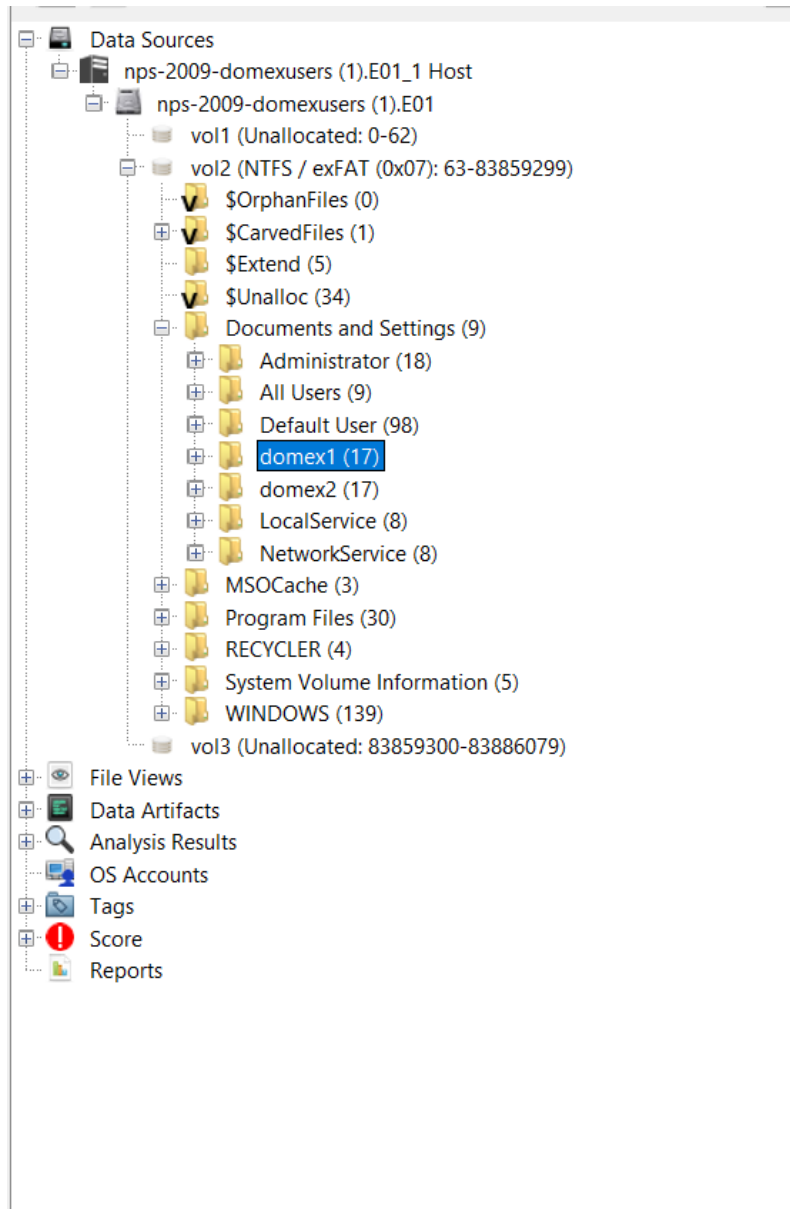


Figure 1. Autopsy Data Sources view showing the Windows user-profile directories, including Administrator, domex1, and domex2.

Artifact 2 – Web History

User: domex2

Source: index.dat web-history artifact

URL:

<https://signup.live.com/mns.aspx?mnsname=domexuser2&mnsdomain=live.com&tk=1225334605114&mkt=en-us&id=64855&ts=4643683&sh=N!Z4&ru=http%3a%2f%2fmail.live.com%2f%3fnewuser%3dyes&rx=http%3a%2f%2fget.live.com%2fmail%2foptions&rollrs=12&lic=1>

Date/Time: 2008-10-30 02:43:25 EDT

What it shows: The web-history artifact associated with the domex2 profile records access to a Live.com signup/mail-related URL. The artifact provides evidence of web activity associated with the domex2 user profile at the recorded timestamp.

Source Name	S	C	O	URL	Date Accessed	Referrer URL	Title
index.dat			1	http://tk2.stb.s-msn.com/i/2E/9CBCCA25F06055268E9FA	2008-10-30 02:42:46 EDT		
index.dat			1	http://get.live.com/	2008-10-30 02:58:40 EDT		
index.dat			1	http://tk2.stb.s-msn.com/i/43/3F8E9A4C42D541F2FD66..	2008-10-30 02:42:46 EDT		
index.dat			1	http://get.live.com/mail/options	2008-10-30 02:43:16 EDT		
index.dat			1	https://signup.live.com/mns.aspx?mnsname=domexuser	2008-10-30 02:43:25 EDT		
index.dat			1	http://m1.2mdn.net/1420759/Imb_iau_MenuSt2quesNEV	2008-10-30 02:44:24 EDT		
index.dat			1	http://gfx8.hotmail.com/mail/13.1.0132.0805/LayoutEdit	2008-10-30 02:57:25 EDT		
index.dat			1	http://tk2.stc.s-msn.com/br/gbl/css/6/decoration/msft....	2008-10-30 02:42:47 EDT		
index.dat			1	http://rad.msn.com/ADSAdClient31.dll?GetSAd=&DPJS.	2008-10-28 20:53:01 EDT		
index.dat			1	http://mail.google.com/support/bin/answer.py?answer...	2008-10-28 20:56:34 EDT		
index.dat			1	http://login.live.com/pp600/WLLogin_JS.srf?x=6.0.11557	2008-10-30 02:42:51 EDT		
index.dat			1	http://b.rad.live.com/ADSAdClient31.dll?GetSAd=&DPJ.	2008-10-30 02:57:23 EDT		
index.dat			1	http://bl135w.blu135.mail.live.com/mail/LayoutEditMes	2008-10-30 02:57:25 EDT		
index.dat			1	http://gfx8.hotmail.com/mail/13.1.0132.0805/LayoutSen	2008-10-30 02:57:30 EDT		
index.dat			1	https://ssl.google-analytics.com/urchin.js	2008-10-28 20:59:21 EDT		

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Result: 127 of 352 Result Web History									
Visit Details Date Accessed: 2008-10-30 02:43:25 EDT Domain: live.com URL: https://signup.live.com/mns.aspx?mnsname=domexuser2&mnsdomain=live.com&tk=1225334605114&mkt=en-us&id=64855&ts=4643683&sh=NIZ4&ru=http%3a%2f%2fmail.l... Program Name: Internet Explorer Analyzer									
Source									

Figure 2. Autopsy Web History result showing the selected Live.com signup/mail-related URL and recorded access time.

Artifact 3 – Deleted File

File Name: adsWrapperAIMAT[1].js

Full Path: /img_nps-2009-domexusers (1).E01/vol_vol2/Documents and Settings/domex1/Local Settings/Temporary Internet Files/Content.IE5/S5IV0D2F/adsWrapperAIMAT[1].js

User: domex1

File Type: JavaScript file

Date/Time: 2008-10-30 03:51:28 EDT

What it shows: The forensic image contains a deleted JavaScript file associated with the domex1 profile and located within the user's Temporary Internet Files directory. The file was identified through forensic examination of the image. A hash was calculated for the exported file using FTK Imager/PowerShell to document the integrity of the examined file.

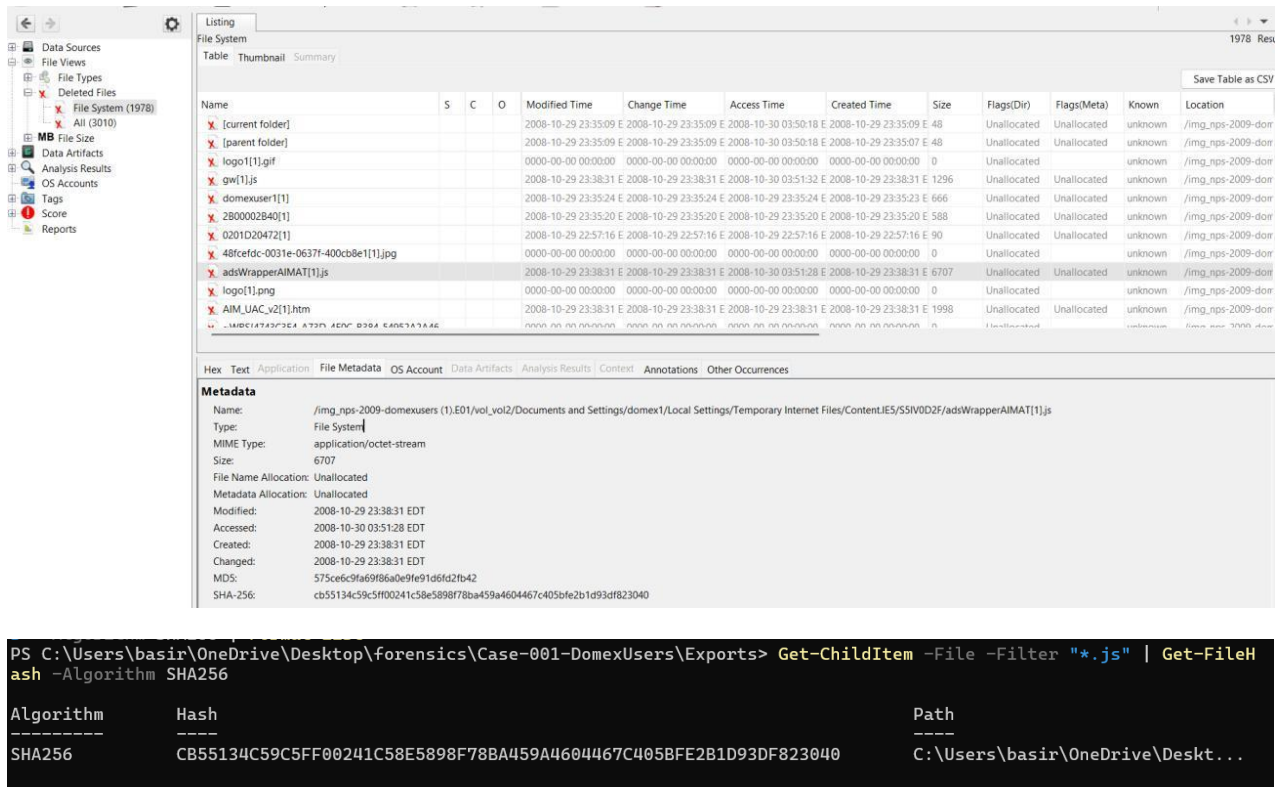


Figure 3. Autopsy deleted-file view showing adsWrapperAIMAT[1].js and associated file metadata, including the SHA-256 value.

Figure 4. PowerShell output showing the SHA-256 hash calculated for the exported JavaScript file.

Artifact 4 – Timeline Event

Date: 2008-10-29

Time: 22:46:00

User: domex2

Event Type: Email Received

Source: Outlook.pst

Path: /img_nps-2009-domexusers (1).E01/vol_vol2/Documents and Settings/domex2/Local Settings/Application Data/Microsoft/Outlook/Outlook.pst

What it shows: The Autopsy timeline identifies an email-received event associated with the domex2 profile. The event is linked to the user's Outlook PST file and provides a timestamp for email activity on October 29, 2008 at 22:46:00 EDT.

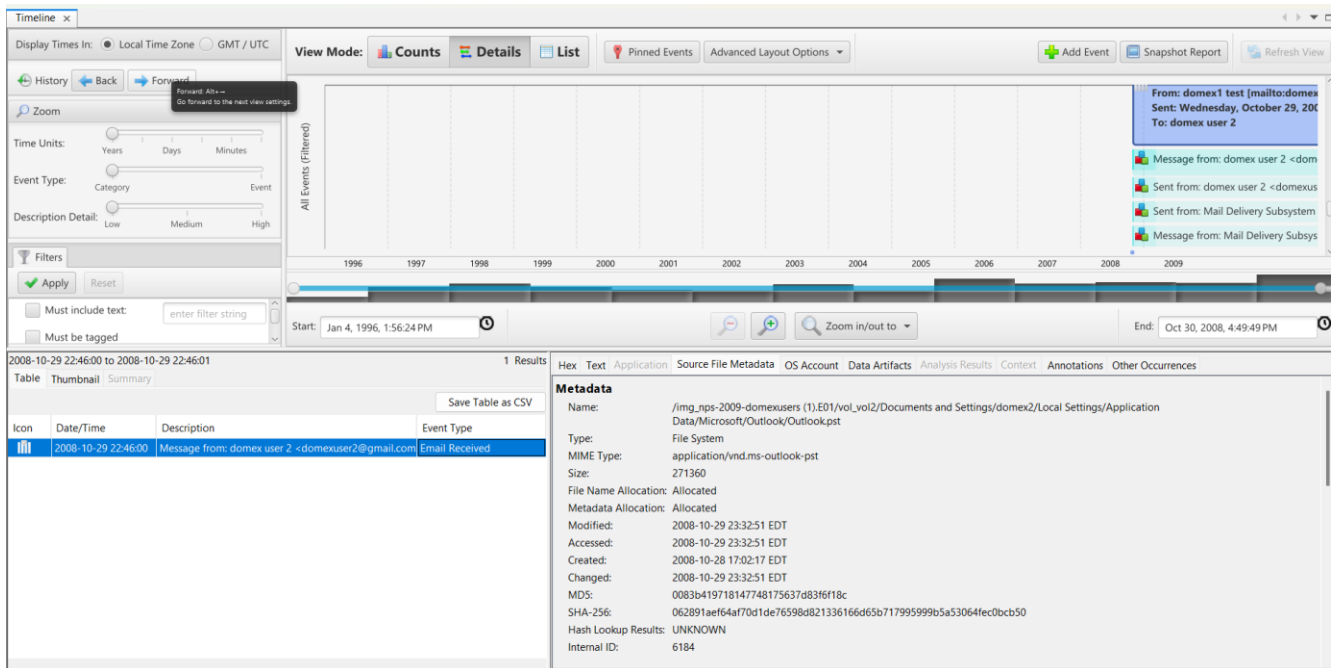


Figure 5. Autopsy Timeline showing an email-received event associated with domex2 on October 29, 2008 at 22:46:00.

Artifact 5 – Windows Registry

Account: domex1

Account Type: Local Windows account

SID: S-1-5-21-842925246-725345543-1844994965-1003

Home Directory: /Documents and Settings/domex1

Account 2: domex2

Account Type: Local Windows account

SID: S-1-5-21-842925246-725345543-1844994965-1004

Home Directory: /Documents and Settings/domex2

Account 3: administrator

Account Type: Local Windows account

SID: S-1-5-21-842925246-725345543-1844994965-500

Home Directory: /Documents and Settings/Administrator

What it shows: The Windows account artifacts establish that administrator, domex1, and domex2 are separate local accounts on the examined Windows system. The unique Security Identifiers (SIDs) distinguish the accounts and provide a basis for correlating account-specific artifacts with their respective user profiles.

Listing

6 Res

TableThumbnailSummary

Save Table as CSV

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-21-842925246-725345543-1844994965-500			0	administrator	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-21-842925246-725345543-1844994965-1003			0	domex1	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-21-842925246-725345543-1844994965-1004			0	domex2	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-19				LOCAL SERVICE	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	
S-1-5-20				NETWORK SERVICE	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	
S-1-5-18				SYSTEM	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Basic Properties

Login: administrator
Full Name:
Address: S-1-5-21-842925246-725345543-1844994965-500
Type:
Creation Date:
Object ID: 7

nps-2009-domexusers (1).E01_1 Host Details

Home Directory: /Documents and Settings/Administrator

Realm Properties

Name: Unknown

Listing

6 Res

TableThumbnailSummary

Save Table as CSV

Name	S	C	O	Login Name	Host	Scope	Realm Name	Creation Time
S-1-5-21-842925246-725345543-1844994965-500			0	administrator	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-21-842925246-725345543-1844994965-1003			0	domex1	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-21-842925246-725345543-1844994965-1004			0	domex2	nps-2009-domexusers (1).E01_1 Host	Local		
S-1-5-19				LOCAL SERVICE	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	
S-1-5-20				NETWORK SERVICE	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	
S-1-5-18				SYSTEM	nps-2009-domexusers (1).E01_1 Host	Local	NT AUTHORITY	

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Basic Properties

Login: domex2
Full Name:
Address: S-1-5-21-842925246-725345543-1844994965-1004
Type:
Creation Date:
Object ID: 3015

nps-2009-domexusers (1).E01_1 Host Details

Home Directory: /Documents and Settings/domex2

Realm Properties

Name: Unknown

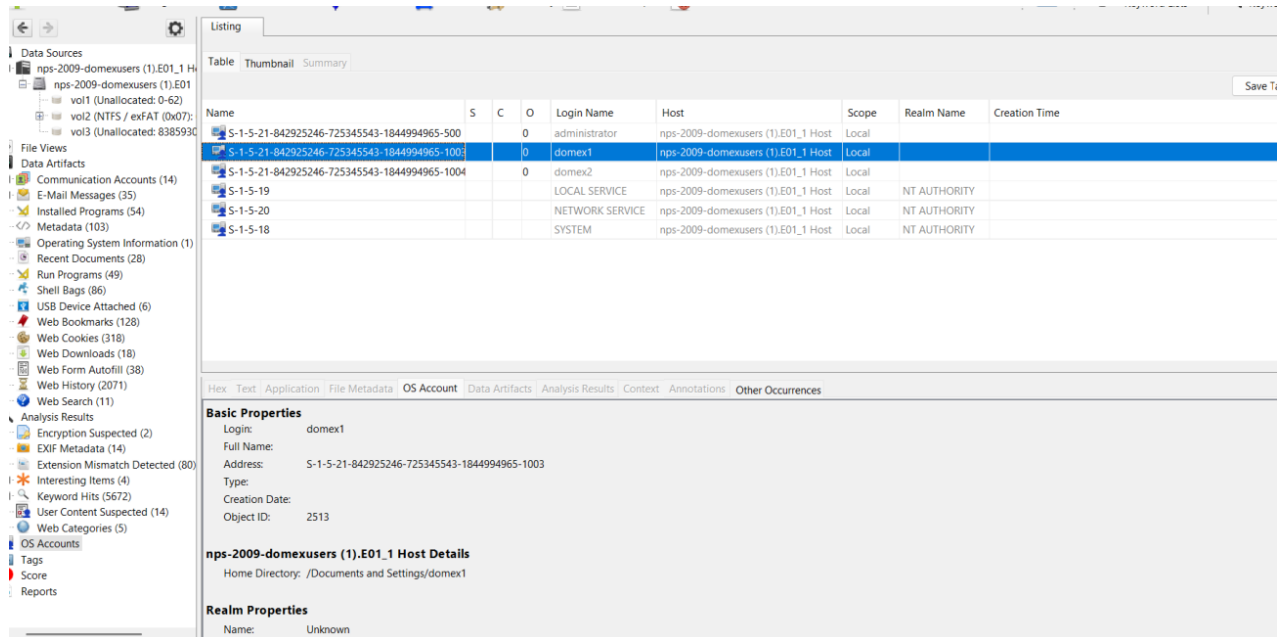


Figure 6. Autopsy OS Account artifact showing the administrator account and its SID/home-directory details.

Figure 7. Autopsy OS Account artifact showing the domex2 account and its SID/home-directory details.

Figure 8. Autopsy OS Account view showing domex1 and the associated SID and home directory.

Artifact 6 – Keyword Hit

Keyword: domex

Filename: Dc3.docx

Full Path: /img_nps-2009-domexusers (1).E01/vol_vol2/RECYCLER/S-1-5-21-842925246-725345543-1844994965-1003/Dc3.docx

User: domex1

Date/Time: 2008-10-29 16:15:00 EDT

What it shows: A keyword search for domex identified Dc3.docx within the RECYCLER directory associated with the SID for domex1. This indicates that a document containing the keyword domex was present in a deleted/recycled-file location associated with the domex1 account.

Listing Keyword search 1 - domex x					
Keyword search					
Table Thumbnail Summary					
Save Table as CSV					
Name	Keyword Preview	Location	Modified Time	Change Time	Access Time
index.dat	word document deleted by «domex» user 1.LNK=0This i	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT
This is a spreadsheet by domex user 1.LNK	this is a spreadsheet by «domex» user 1.LNK	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:16:28 EDT	2008-10-29 12:16:28 EDT	2008-10-29 12:16:28 EDT
Dc3.docx	word document deleted by «domex» user 1.LNK=0This i	/img_nps-2009-domexusers (1).E01/vol_vol2/RECYCLER	2008-10-29 12:15:34 EDT	2008-10-29 23:39:32 EDT	2008-10-29 23:39:32 EDT
_REGISTRY_USER_NTUSER_S-1-5-21-842925246-72	a spreadsheet sent by «domex» user 1.LNKThis is a	/img_nps-2009-domexusers (1).E01/vol_vol2/System Vo	2008-10-30 12:47:55 EDT	2008-10-30 12:49:00 EDT	2008-10-30 12:49:00 EDT
This is a spreadsheet by domex user 1.LNK-slack	this is a spreadsheet by «domex» user 1.LNK-slack	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:16:28 EDT	2008-10-29 12:16:28 EDT	2008-10-29 12:16:28 EDT
This is a spreadsheet deleted by domex user 1.LNK	spreadsheet deleted by «domex» user 1.LNK	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT
Dc4.xlsx	spreadsheet deleted by «domex» user 1.LNK=0This i	/img_nps-2009-domexusers (1).E01/vol_vol2/RECYCLER	2008-10-29 12:17:24 EDT	2008-10-29 23:39:38 EDT	2008-10-29 23:39:38 EDT
This is a spreadsheet deleted by domex user 1.LNK	spreadsheet deleted by «domex» user 1.LNK-slack	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT	2008-10-29 12:17:24 EDT
_REGISTRY_USER_NTUSER_S-1-5-21-842925246-72	Use AuthDisplay Name«domex» user 2Emaildomexuser	/img_nps-2009-domexusers (1).E01/vol_vol2/System Vo	2008-10-29 23:00:12 EDT	2008-10-29 23:01:10 EDT	2008-10-29 23:01:10 EDT
This is a spreadsheet sent by domex user 1.LNK	a spreadsheet sent by «domex» user 1.LNK	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:16:52 EDT	2008-10-29 12:16:52 EDT	2008-10-29 12:16:52 EDT
This is a spreadsheet sent by domex user 1.LNK-slack	a spreadsheet sent by «domex» user 1.LNK-slack	/img_nps-2009-domexusers (1).E01/vol_vol2/Document	2008-10-29 12:16:52 EDT	2008-10-29 12:16:52 EDT	2008-10-29 12:16:52 EDT
Dc3.docx	deleted and emptied by «domex» user 1.LNK=0This i	/img_nps-2009-domexusers (1).E01/vol_vol2/RECYCLER	2008-10-29 23:40:07 EDT	2008-10-29 23:40:07 EDT	2008-10-29 23:40:07 EDT

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata									
Name:	/img_nps-2009-domexusers (1).E01/vol_vol2/RECYCLER/S-1-5-21-842925246-725345543-1844994965-1003/Dc3.docx								
Type:	File System								
MIME Type:	application/vnd.openxmlformats-officedocument.wordprocessingml.document								
Size:	9852								
File Name Allocation:	Allocated								
Metadata Allocation:	Allocated								
Modified:	2008-10-29 12:15:34 EDT								
Accessed:	2008-10-29 23:39:30 EDT								
Created:	2008-10-29 12:15:34 EDT								
Changed:	2008-10-29 23:39:32 EDT								
MDS:	53dec327aed600884f2145fc860aaee8								
SHA-256:	bff5f32a373c59fc9ca83df70d5b2267ec8a327b934d5e7fdac786d8f75d894								

Figure 9. Autopsy keyword-search result for “domex” identifying Dc3.docx in the RECYCLER path associated with the domex1 SID.

Correlation

1. 2008-10-29 16:15:00 EDT — domex1

A keyword search identified Dc3.docx in the RECYCLER directory associated with the domex1 SID.

2. 2008-10-29 22:46:00 EDT — domex2

The Autopsy timeline records an email-received event associated with the domex2 Outlook PST.

3. 2008-10-30 02:43:25 EDT — domex2

Web-history evidence records access to a Live.com signup/mail-related URL.

4. 2008-10-30 03:51:28 EDT — domex1

A deleted JavaScript file, adsWrapperAIMAT[1].js, is identified in the domex1 Temporary Internet Files directory.

Relationship between artifacts: The artifacts demonstrate activity associated with two separate user profiles, domex1 and domex2, during a relatively short period from October 29–30, 2008. The evidence shows a chronological sequence involving a document located in a recycled-file location associated with domex1, an email-received event associated with domex2, subsequent Live.com web activity associated with domex2, and a deleted JavaScript file associated with domex1. These artifacts can be correlated temporally and by user profile; however, the available artifacts alone do not establish that all events were performed by the same individual or that the events were causally related.

Limitations

The analysis is based on artifacts recovered from the provided forensic image and the timestamps and metadata generated by the forensic tools. The presence of a user account or profile does not independently establish the identity of the person operating the computer. Web-history, email, deleted-file, and keyword

artifacts provide evidence of activity but may not establish the intent of the user. Timestamps may also be affected by system time configuration or other factors. Additional examination of registry artifacts, email contents, web artifacts, file metadata, deleted files, and timeline events would be required to establish stronger relationships between the observed activities.